

# Exact Event Maps and Cluster Coarsening in Pulse-Coupled Integrate-and-Fire Networks

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## Abstract

We give a finite exact event certificate for an all-to-all excitatory network of identical integrate-and-fire oscillators. The strictly concave rise  $U_a(\phi) = (1 - e^{-a\phi})/(1 - e^{-a})$  becomes rational event arithmetic in  $u = e^{-a\phi}$  when  $r = e^{-a}$  is rational. Free flight is a common rescaling, a pulse subtracts  $(1 - r)\epsilon$  with threshold clipping, and same-time avalanche closure is explicit. We prove equality-partition coarsening and the absorbing synchronized primitive event cycle  $[N]$ . The receipt contains 441 finite probes for  $N \leq 8$ , an independent checker, symbolic checks, replay, and hostile mutations. Mirollo–Strogatz synchrony is cited separately; no exhaustive continuous-state census or target spectral claim is made. The strict Route-A verdict is `ROUTE_A_REJECTED`.

## 1 Model and event coordinates

Let  $N \geq 2$  phases satisfy  $\dot{\phi}_i = 1$  on  $[0, 1)$ . With  $r = e^{-a} \in (0, 1)$ , define

$$U_a(\phi) = \frac{1 - e^{-a\phi}}{1 - r}, \quad u_i = e^{-a\phi_i}. \quad (1)$$

Threshold  $U_a = 1$  is  $u = r$ , and reset is  $U_a = 0$ ,  $u = 1$ . A firing oscillator sends an excitatory increment  $\epsilon \in (0, 1)$  to every nonfiring oscillator in the  $U$ -coordinate. Pulses are closed at the same time: a newly thresholded oscillator fires and sends one further pulse.

**Theorem 1** (Exact event map). *For a post-avalanche state  $u \in [r, 1]^N$ , free flight to the next event is*

$$\hat{u}_i = \frac{r}{\min_j u_j} u_i. \quad (2)$$

*The threshold set is  $F_0 = \{i : \hat{u}_i = r\}$ . If  $F_k$  is one simultaneous recruitment wave, every nonfiring coordinate is updated by*

$$u_j \leftarrow u_j - (1 - r)\epsilon |F_k|, \quad F_{k+1} = \{j \notin F_0 \cup \dots \cup F_k : u_j \leq r\}, \quad (3)$$

*then all indices in the finite closure  $F = \bigcup_k F_k$  reset to one.*

*Proof.* During free flow  $u_i(t) = u_i(0)e^{-at}$ ; solving  $e^{-at} \min_i u_i = r$  gives (2). Since  $U_a^{-1}(y) = \log(1 - (1 - r)y)/\log r$ , adding  $\epsilon$  in  $y$  subtracts  $(1 - r)\epsilon$  in  $u$ , with clipping at  $r$ . Applying all pulses in a wave simultaneously yields (3). Each wave adds at least one new index, so the closure terminates in at most  $N$  waves.  $\square$

**Proposition 1** (Rationality and no splitting). *If  $r, \epsilon$ , and the event state are rational, every event state is rational. Moreover, if two labelled coordinates are equal before an event, they are equal after it; equivalently every old equality block is contained in a unique new equality block.*

*Proof.* All operations in (2)–(3) are rational comparisons and additions/multiplications. Common scaling preserves equality. A nonfiring equality block receives the same subtraction in every wave; if it reaches threshold, the whole block is recruited and reset together. Thus no old block can split, even though different blocks may merge during an avalanche.  $\square$

## 2 Synchronized cycle and finite certificate

At  $u = (1, \dots, 1)$ , (2) gives  $u = (r, \dots, r)$ ; all  $N$  indices fire, and reset returns to  $(1, \dots, 1)$ . The event word is the one-letter word  $[N]$ , hence primitive of period one. This is a source-local event cycle, not an arithmetic primitive orbit.

The exact ledger uses  $r \in \{1/2, 2/3, 3/4\}$ ,  $\epsilon \in \{1/5, 1/4, 1/3\}$ ,  $N = 2, \dots, 8$ , seven deterministic rational seeds, and twelve event steps: 441 rows. It retains 63 all-equal synchronized rows and records every recruitment wave, old-block containment receipt, and cycle label. It intentionally does not enumerate all continuous parameter cells or all initial states.

Table 1: Representative exact event rows. The event word lists firing cardinalities; recruitment waves are serialized separately in the receipt.

| $r$ | $\epsilon$ | $N$ | seed | event word            | cycle |
|-----|------------|-----|------|-----------------------|-------|
| 1/2 | 1/5        | 2   | 0    | [2]                   | 1     |
| 2/3 | 1/4        | 4   | 0    | [4]                   | 1     |
| 3/4 | 1/3        | 8   | 0    | [8]                   | 1     |
| 1/2 | 1/5        | 2   | 1    | [1,1,1,1,1,1,1,1,1,1] | –     |

## References

- [1] R. E. Mirollo and S. H. Strogatz, “Synchronization of Pulse-Coupled Biological Oscillators,” *SIAM Journal on Applied Mathematics* 50 (1990), 1645–1662, DOI: 10.1137/0150098.
- [2] S. Bottani, “Synchronization of integrate and fire oscillators with global coupling,” *Physical Review E* 54 (1996), 2334–2340, DOI: 10.1103/PhysRevE.54.2334.

## Declarations

**Scope.** NO\_BAD\_EULER\_OR\_ROOT\_NUMBER; no arithmetic, target-zero, Euler-factor, root-data, automorphy, or Hilbert–Pólya claim is made. **Data and code.** The exact receipt, independent checker, replay, and mutation audit are included. **AI-use disclosure.** Generative tools assisted drafting and code generation; formulas and metadata are checked by the deterministic artifact chain.